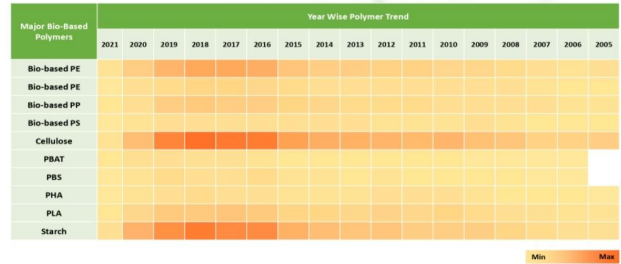


My Illustrative Infographic Flyers

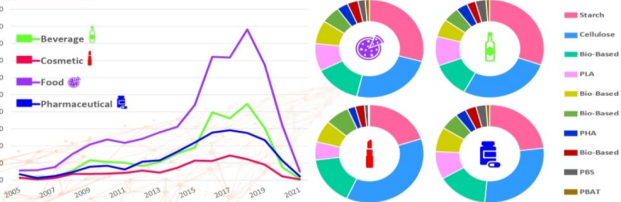
Green Alternatives For Bio-Based Polymers

Exploring the Evolution of Bio-Based Polymers



Majority of players have focused upon **Cellulose & Starch** over the years in terms of IP activity. It has numerous advantages over the other polymers like it offers safety to the environment and can be reused, recycled, and renewed. It is harvested from plant materials, such as byproducts of agricultural processing and production.

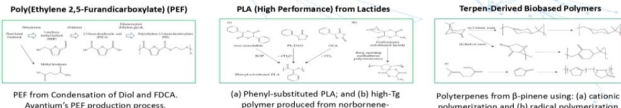
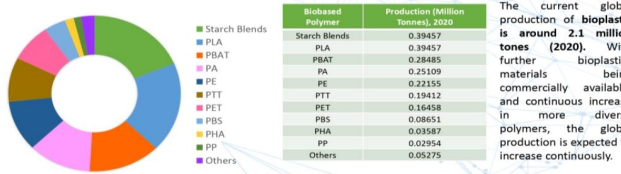
Unveiling the Latest Trends in Bio-Based Polymer Applications



Conquering Thermal Stability Challenges in Polymer Innovation

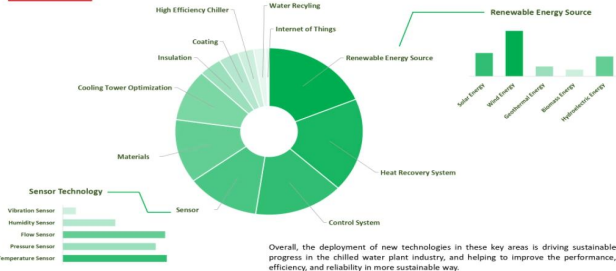
- Poly(lactic Acid) Plastics**
 - Low molecular weight compounds such as oligomeric lactic acid, glycerol, triacetin, and low molecular weight citrates, partial fatty acid esters are common plasticizers of PLA
- Polyhydroxyalkanoates (PHAs)**
 - Use of monomeric and polymeric plasticizers of PHA lowers the glass transition temperature and the melting points, allowing processing at lower temperature and avoiding thermal degradation
 - The plasticization of PHBV using soybean oil, epoxidized soybean oil, epoxidized linseed oil, dibutyl phthalate, polyester plasticizer (Lapal 108), triethyl citrate, acetyl tributyl citrate and polyethylene glycol can be useful
- Thermoplastic Starch (TPS)**
 - Plasticizers for the preparation of thermoplastic starch are glycerol, glycol, sorbitol, xylitol, maltitol, ethylene glycol, propylene glycol, butanediol, sucrose, fructose, mannose, fatty acids (such as myristate or palmitate), etc

Breaking Down Global Production Capacity



Chilled Water Plant Sustainable Technology

Water Treatment Plant Sustainable Technology Segmentation



Competitors Sustainable Approaches in Chilled Water Plant

- Hilton**
 - Hilton has implemented high-efficiency chillers, heat recovery systems, and advanced control systems in many of its properties worldwide. As a result, the company has been able to reduce energy consumption by up to 20%, lower water usage, and reduce greenhouse gas emissions.
- Carrier**
 - Carrier is a leading provider of HVAC systems and services, offering a range of sustainable solutions for chilled water plants. Their products include energy-efficient chillers, cooling towers, and controls, as well as renewable energy systems such as solar and geothermal. Carrier's solutions are designed to reduce energy consumption and carbon footprint while improving comfort and indoor air quality.
- Johnson Controls**
 - Johnson Controls has developed advanced cooling solutions that use high-efficiency chillers, heat recovery systems, and advanced control systems to reduce energy consumption and lower environmental impact. The company's systems have been installed in many commercial and industrial buildings worldwide.
- Schneider Electric**
 - Schneider Electric has developed an advanced cooling system called EcoStruxure, which uses free cooling, heat recovery, and advanced control systems to achieve energy savings of up to 50% compared to traditional cooling systems.

How Renewable Energy Helping Industries To Achieve Sustainability



PV Modules, Solar Energy and Wastewater Management are the key sustainable technology focus of the industries. Recently in 2019, Facebook announced that it had installed a 2.2 MW solar panel system at its data center in Singapore. The solar panels generate approximately 1.2 million kWh of electricity per year, which is used to power the data center's chilled water system. This has helped Facebook to reduce its carbon footprint and increase its use of renewable energy.

Automation And Real-Time Monitoring Is The Future !

Location: San Francisco Museum of Modern Art, San Francisco, CA 94103

Problem: The museum, which has a total floor area of 460,000 square feet, required a cooling system that could provide optimal temperature and humidity control while reducing energy consumption and greenhouse gas emissions.

Solution:

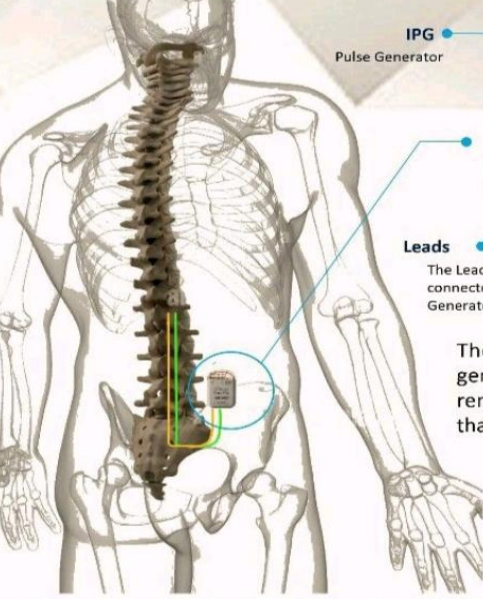
The retrofit project involved the replacement of the existing chillers with a new, energy-efficient system consisting of two 550-ton centrifugal chillers with magnetic bearing compressors. The new system also included a thermal energy storage (TES) tank with a capacity of 200,000 gallons, which enabled the plant to shift energy consumption from peak to off-peak hours and reduce electricity demand charges.

In addition, the project included the installation of a new control system with advanced features such as fault detection and diagnostics, which allowed for real-time monitoring and optimization of the plant's performance. The retrofit was completed in 2019 and has since reduced energy consumption by 2.5 million kilowatt-hours per year, resulting in an annual reduction of 1,500 metric tons of carbon dioxide emissions.

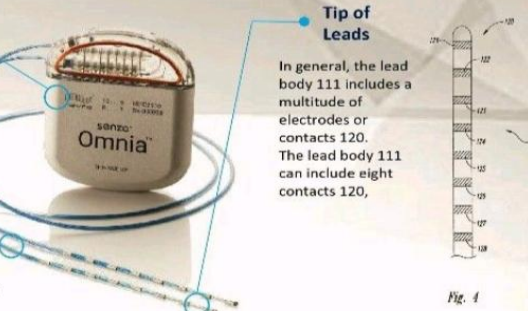
Conclusion:

Overall, the retrofit of the SFMOMA cooling system is an excellent example of how advanced technologies and innovative design can be used to achieve significant reductions in carbon emissions in chilled water plants. It demonstrates the potential of energy-efficient systems, thermal energy storage, and advanced controls to deliver sustainable and cost-effective cooling solutions for large commercial buildings.

Spinal Cord Stimulation System



The Senza Spinal Cord Stimulation (SCS) system is an implanted, rechargeable spinal cord stimulation device designed to relieve persistent, difficult-to-manage pain in the trunk or limbs.

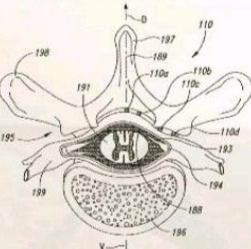


The primary parts of the SCS system are an implanted signal generator that is coupled to one or two implanted leads and a remote control that can activate or deactivate the stimulator so that therapeutic parameters can be changed.

The Senza SCS systems' indications for use are expanded by this supplement to address lower limb pain caused by diabetic neuropathy, sometimes known as painful diabetic neuropathy.

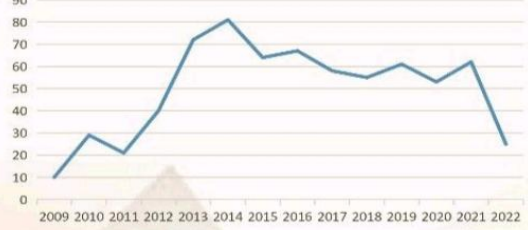
The remote control sends radio signals to the implanted signal generator, instructing it when to provide the proper stimulation to the spinal cord. The patient or a healthcare professional can operate the battery-powered external remote control. The SCS system when programmed to a frequency of 10 kHz, may help cure chronic, intractable pain in the patient's low back, trunk, or limbs.

FIG. 6 is a cross-sectional illustration From US '343 A1



- Spinal Cord 191
- Adjacent Vertebra 195
- Lead Bodies 110 a-110 d
- Ventral Body 196
- Vertebra 195
- Dura Mater 199
- Dorsal Roots 193
- Dorsal Root Ganglia 194

To send signals directly to the spinal cord 191, a lead body 110 can be positioned centrally in a lateral direction, for example, aligned with the midline of the spinal cord 189. To give the Spinal Cord 191 both a paresthetic and anaesthetic effect, electrical pulses may be delivered to the lead body 110.



The Technology Filing Trend shows that from 2009, Nevro has been continuously working to improve the SCS system. The FDA recently approved Nevro Corporation's SCS System in June 2022.